

Validity Analysis: GSM8K

Target context: US K-8 Math Tutoring Platform — Homework Checker Deployment

Overall risk: **MEDIUM**

Deployment Context

We are building a homework-checking feature for a US K-8 math tutoring platform. Students type in or receive arithmetic word problems through the platform's text interface, and our LLM checks whether their final numerical answer is correct. We want to evaluate the LLM's ability to solve multi-step arithmetic word problems (addition, subtraction, multiplication, division, fractions, percentages) before deploying this feature to American elementary and middle school students.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	3	Moderate gaps	high
Input Content ■	4	Minor gaps	high
Input Form ✓	5	Strong alignment	high
Output Ontology ✓	5	Strong alignment	high
Output Content	4	Minor gaps	high
Output Form ✓	5	Strong alignment	high
Average	4.3		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

GSM8K is well-suited as a screening-tier benchmark for the platform's core use case — multi-step arithmetic word problem checking for grades 3-6 students using US conventions — but has confirmed coverage gaps for several content types the platform serves regularly. The strongest alignment is on output structure (binary final-answer checking is structurally identical), input form (text-only English), and cultural framing (USD currency and American settings dominate empirically). The weakest alignment is in input ontology: US customary unit conversion problems were not observed in 80 sampled examples and are not documented in the paper; basic statistics (mean/median) is confirmed absent by multiple web sources and the sample; area/perimeter problems are confirmed absent (volume and area-as-pricing present but no perimeter); K-2 single-step arithmetic is excluded by the 2-step minimum design floor. These four gaps mean that strong GSM8K

performance does not guarantee competence on roughly the lower-grade (K-2) and upper-grade-supplementary (statistics, perimeter, conversion) portions of the platform's content inventory. Additionally, benchmark saturation at ~95%+ for frontier models [WEB-4] further limits discrimination at the top end. For the platform's dominant grade 3-6 multi-step four-operations use case, GSM8K provides solid evaluation evidence; for the documented gap areas, supplementary evaluation is required.

ID	Evidence
WEB-4	https://arxiv.org/pdf/2509.25160

Practical Guidance

What This Benchmark Measures

GSM8K validly measures an LLM's ability to read American English arithmetic word problems and produce correct final numerical answers via multi-step reasoning involving the four operations, fractions, decimals, percentages, ratios, and rates — the platform's highest-volume grade 3-6 content (input_ontology, input_content, output_ontology, output_form all score 3-5). It also confirms baseline cultural framing (USD, US customary units, American settings) appropriate for the target population.

Construct Depth

Depth is high for multi-step four-operations arithmetic but shallow or absent for several CCSS-M content types the platform requires: US customary unit conversion (in↔ft, oz↔lb, cups↔gallons), basic statistics (mean/median from datasets), perimeter geometry, and K-2 single-step arithmetic. The benchmark provides strong signal on the platform's dominant use case (~60-70% of platform volume by elicited grade band) but limited signal on the remaining sub-domains. Saturation at ~95%+ further reduces discrimination among frontier models [WEB-4].

What Else You Need

(1) A CCSS-M-aligned measurement conversion test set to fill the input_ontology gap on US customary unit conversion. (2) A grades 6-8 statistics word problem set (mean/median/mode/range from small datasets) — fully absent per [WEB-5, WEB-7] and sample. (3) A perimeter-focused geometry word problem set to complement GSM8K's area/volume coverage. (4) A K-2 single-step arithmetic test set, as GSM8K's 2-step minimum excludes this band by design. (5) An ambiguity audit on the deployment-relevant subset to mitigate the documented 1.7% label-error rate impact on user-facing scoring.

ID	Evidence
WEB-4	https://arxiv.org/pdf/2509.25160
WEB-5	Recent cultural adaptation work (Tomar et al. 2025) created non-US variants, confirming original is implicitly US/Western-anchored (https://www.emergentmind.com/topics/gsm8k-dataset-95a37205-4213-4591-900f-09ce82d0a123)
WEB-7	'no advanced mathematics (no geometry, quadratics, combinatorics)'; no statistics mentioned (https://mbrenndoerfer.com/writing/gsm8k-evaluating-mathematical-reasoning-language-models)

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: GSM8K | Context: US K-8 Math Tutoring Platform — Homework Checker Deployment

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K's core taxonomy — multi-step arithmetic word problems using the four operations with fractions, percentages, and ratios — directly matches the platform's highest-volume grade 3-6 use case [\[Q1, Q11, Q19\]](#) and is empirically confirmed across the dataset sample [\[DATASET-D13, D17, D18, D21, D24\]](#). However, three platform-relevant categories are documented as absent or sparse: (a) US customary unit conversion problems are not documented in the paper and no conversion problem appeared in 80 sampled examples ([DATASET-D31](#), D32 show only time/rate conversions, not in↔ft or oz↔lb); (b) basic statistics (mean/median from datasets) is confirmed absent in web research [\[WEB-5, WEB-7\]](#) and only a single embedded mean problem appeared in the sample [\[DATASET-D10\]](#); (c) area/perimeter geometry is confirmed absent per [\[WEB-5, WEB-7\]](#), with the sample showing volume and area-as-pricing [\[DATASET-D11, D12\]](#) but no perimeter problems. Additionally, the benchmark's 2–8 step difficulty floor [\[WEB-5, WEB-6\]](#) excludes true K-2 single-step arithmetic by design, leaving the platform's K-2 cohort untested. The benchmark also includes some algebra-required problems [\[DATASET-D16, D22\]](#) that exceed the platform's K-6 arithmetic scope, introducing minor construct-irrelevant variance. Given MODERATE priority and confirmed gaps on 3 of the user's explicitly flagged content types, partial alignment is the most defensible score.

ID	Evidence
Q1	"we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems." (p.1)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
Q19	"However, we note that GSM8K is larger, provides natural language solutions, and consists of problems that on average require more steps to solve." (p.4)
D10	calculate the average age of the three?" — solved as "105 years / 3 people = <<105/3=35>>35 years/person — Mean/average calculation embedded in a word problem — statistical reasoning present
D11	4 feet long, 6 feet wide, and 3 feet high... 4 ft * 6 ft * 3 ft = <<4*6*3=72>>72 cubic ft... 72 cubic ft * 3/4 = <<72*3/4=54>>54 cubic ft — Volume calculation using length × width × height — geometry-adjacent area/volume
D12	The price of a home is \$98 per square foot (sq ft). The house is 2,400 sq ft and the barn out back is 1,000 sq ft. How much is this property? — Area (sq ft) applied to pricing — geometry-adjacent; uses sq ft (US customary)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Multi-step four-operations problem across sequential days
D16	Jim has 4 times as many Buicks as Fords, and 3 more than twice the number of Fords than Chevys — Algebraic reasoning with US car brands — requires variable setup
D17	The bottle of milk cost was 75% of the total cost of the sandwich and juice" — "75/100 * 12 = \$<<75/100*12=9>>9 — Percentage calculation embedded in word problem
D18	her rent goes up by 30%, food costs increase by 50%, and the cost of her car insurance triples — Multi-step percentage increase problem with monthly/annual scaling
D21	She gives away 4/5 of the cupcakes to her classmates. Of the remaining 1/5 of cupcakes, she eats 3 cupcakes. — Fraction operations on a whole — grades 4-5 fraction content
D22	Ten friends decide to get an end-of-year gift for their teacher... four of the group drop out. The remaining friends split the cost equally... each share is now \$8 more — Algebraic reasoning with cost-splitting
D24	Each bird eats 12 beetles per day, each snake eats 3 birds per day, and each jaguar eats 5 snakes per day. If there are 6 jaguars in a forest — Multi-step chained multiplication — nature/ecology context

D31	Larry spends half an hour twice a day walking and playing with his dog. He also spends a fifth of an hour every day feeding his dog. How many minutes does Larry spend on his dog each day? — Unit conversion (hours to minutes) within problem — time measurement conversion
D32	Roberto can skip 4,200 times an hour. Valerie can skip 80 times a minute. If they jump rope for fifteen minutes straight — Rate conversion (hours to minutes) required for solution
WEB-5	Recent cultural adaptation work (Tomar et al. 2025) created non-US variants, confirming original is implicitly US/Western-anchored (https://www.emergentmind.com/topics/gsm8k-dataset-95a37205-4213-4591-900f-09ce82d0a123)
WEB-6	Solutions 'primarily involve performing a sequence of elementary calculations using basic arithmetic operations' (https://huggingface.co/datasets/openai/gsm8k)
WEB-7	'no advanced mathematics (no geometry, quadratics, combinatorics)'; no statistics mentioned (https://mbrenndoerfer.com/writing/gsm8k-evaluating-mathematical-reasoning-language-models)

Strengths

- Core multi-step four-operations arithmetic at grades 3-6 difficulty is well represented ([DATASET-D13](#), [D24](#); [Q19](#))
- Fractions and percentages are present and match grades 4-6 content [[DATASET-D17](#), [D18](#), [D21](#), [D26](#)]
- Ratio and proportional reasoning problems cover grades 6-8 content [[DATASET-D14](#), [D15](#)]
- High linguistic diversity in problem framing reduces template-overfitting risk [[Q9](#), [Q10](#), [Q113](#)]

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q10	"State-of-the-art language models struggle to achieve high performance on this dataset, primarily due to the high diversity among problems." (p.2)
Q19	"However, we note that GSM8K is larger, provides natural language solutions, and consists of problems that on average require more steps to solve." (p.4)
Q113	"To ensure contractors were not re-using problem templates, we computed pairwise similarity scores between problems and used this to provide feedback to contractors." (p.15)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Multi-step four-operations problem across sequential days
D14	There are enough provisions in a castle to feed 300 people for 90 days. After 30 days, 100 people leave the castle. How many more days are left until all the food runs out? — Multi-step proportional reasoning (rates/ratios) — castle setting, not US-specific
D15	The ratio of coins that Elsa has to that which Amalie has is 10:45. If the total number of coins they have is 440, and Amalie spends 3/4 of what she has on toys — Ratio problem with non-US names; no currency denomination specified
D17	The bottle of milk cost was 75% of the total cost of the sandwich and juice" — " $75/100 * 12 = \$ << 75/100 * 12 = 9 >> 9$ — Percentage calculation embedded in word problem
D18	her rent goes up by 30%, food costs increase by 50%, and the cost of her car insurance triples — Multi-step percentage increase problem with monthly/annual scaling
D21	She gives away 4/5 of the cupcakes to her classmates. Of the remaining 1/5 of cupcakes, she eats 3 cupcakes. — Fraction operations on a whole — grades 4-5 fraction content
D24	Each bird eats 12 beetles per day, each snake eats 3 birds per day, and each jaguar eats 5 snakes per day. If there are 6 jaguars in a forest — Multi-step chained multiplication — nature/ecology context
D26	He has gotten 8 haircuts and knows that he needs 2 more to reach his goal. What percentage towards his goal is he? — Percentage of goal completion — straightforward percentage

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories

IO-4	Document category gaps harming content validity
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Checklist responses:

IO-1: Required categories per elicitation: multi-step four-operations arithmetic (grades 3-6), fractions/decimals/percentages/ratios (grades 4-8), US customary unit conversion, basic statistics (mean/median), area/perimeter word problems, and K-2 single-step arithmetic.

IO-2: Yes. Three categories are omitted or sparsely represented: (a) US customary unit conversion — not documented in paper; absent from 80-example sample ([DATASET-D31](#), D32 only show time conversions); (b) basic statistics (mean/median) — confirmed absent per [\[WEB-5, WEB-7\]](#); only one embedded mean problem in sample [\[DATASET-D10\]](#); (c) area/perimeter — confirmed absent per [\[WEB-5, WEB-7\]](#); sample shows volume/area-as-pricing only [\[DATASET-D11, D12\]](#), no perimeter. Additionally, K-2 single-step arithmetic is excluded by the 2-step minimum design floor [\[WEB-5, WEB-6\]](#).

IO-3: Minor inclusion of algebra-required variable-equation problems [\[DATASET-D16, D22\]](#) that exceed the platform's stated K-8 arithmetic scope. Low-frequency but introduces some construct-irrelevant variance.

IO-4: Documented gaps that harm content validity: K-2 single-step problems (full gap, by design [\[WEB-5, WEB-6\]](#)); statistics word problems (full gap [\[WEB-5, WEB-7\]](#)); perimeter problems (full gap [\[WEB-5, WEB-7\]](#)); US customary unit conversion (likely partial gap, unconfirmed at full-dataset scale). Strong coverage on core multi-step four-operations and fractions/percentages/ratios.

ID	Evidence
Q1	"we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems." (p.1)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
Q48	"Assuming a log-linear trend, we can naively extrapolate these results to estimate that a model with 10^{16} parameters would be required to reach an 80% solve rate, when using the full GSM8K training set." (p.6)
D10	calculate the average age of the three?" — solved as "105 years / 3 people = $\ll 105/3=35 \gg$ 35 years/person — Mean/average calculation embedded in a word problem — statistical reasoning present
D11	4 feet long, 6 feet wide, and 3 feet high... $4\text{ ft} \times 6\text{ ft} \times 3\text{ ft} = \ll 4*6*3=72 \gg$ 72 cubic ft... $72\text{ cubic ft} \times 3/4 = \ll 72*3/4=54 \gg$ 54 cubic ft — Volume calculation using length $\times$ width $\times$ height — geometry-adjacent area/volume
D12	The price of a home is \$98 per square foot (sq ft). The house is 2,400 sq ft and the barn out back is 1,000 sq ft. How much is this property? — Area (sq ft) applied to pricing — geometry-adjacent; uses sq ft (US customary)
D16	Jim has 4 times as many Buicks as Fords, and 3 more than twice the number of Fords than Chevys — Algebraic reasoning with US car brands — requires variable setup
D22	Ten friends decide to get an end-of-year gift for their teacher... four of the group drop out. The remaining friends split the cost equally... each share is now \$8 more — Algebraic reasoning with cost-splitting
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WEB-5	Recent cultural adaptation work (Tomar et al. 2025) created non-US variants, confirming original is implicitly US/Western-anchored (https://www.emergentmind.com/topics/gsm8k-dataset-95a37205-4213-4591-900f-09ce82d0a123)
WEB-6	Solutions 'primarily involve performing a sequence of elementary calculations using basic arithmetic operations' (https://huggingface.co/datasets/openai/gsm8k)
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Information Gaps

- Frequency of US customary unit conversion problems in the full 8,500-problem corpus (sample of 80 found none, but small sample limits inference)

- Whether any median/range/mode problems exist anywhere in the full dataset
- Whether GSM8K problems have any incidental alignment to specific CCSS-M grade-level standards

Requires Expert Verification

- Full-dataset frequency audit of measurement conversion, statistics, and geometry sub-skills via keyword/semantic search
- Mapping of GSM8K items to CCSS-M grade bands by a curriculum specialist

Remediation

Gap 1: US customary unit conversion problems (feet↔inches, cups↔gallons, pounds↔ounces) not observed in 80-example sample and not documented in the paper; CCSS-M-mandated for grades 4-5+

Recommendation: Construct a 100-200 problem supplementary test set of US customary unit conversion word problems aligned to CCSS-M grades 4-5 measurement standards. Evaluate the LLM on this set alongside GSM8K.

Gap 2: Basic statistics word problems (mean, median, mode, range from small datasets) are confirmed absent from GSM8K's documented content scope [WEB-5, WEB-7]; only one embedded mean problem in 80-example sample

Recommendation: Add a grades 6-8 statistics word problem evaluation set covering mean/median/mode/range from named small datasets to fill this full coverage gap before deploying for upper-grade students.

Gap 3: Perimeter word problems absent from sample; geometry confirmed not in GSM8K scope [WEB-5, WEB-7]

Recommendation: Supplement with a perimeter-focused word problem set for rectangles, composite shapes, and real-world contexts. GSM8K's volume/area-as-pricing problems do not substitute for perimeter reasoning.

Gap 4: K-2 single-step arithmetic is excluded by GSM8K's 2-step minimum design floor [WEB-5, WEB-6]; deployment serves K-2 students

Recommendation: Evaluate K-2 capability with a single-step single-operation word problem set if the platform applies the homework-checker to early-grade student submissions. GSM8K provides zero test coverage for this band.

ID	Evidence
WEB-5	Recent cultural adaptation work (Tomar et al. 2025) created non-US variants, confirming original is implicitly US/Western-anchored (https://www.emergentmind.com/topics/gsm8k-dataset-95a37205-4213-4591-900f-09ce82d0a123)
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Input Content (IC)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: GSM8K | Context: US K-8 Math Tutoring Platform — Homework Checker Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Empirical sampling provides strong evidence that GSM8K's content predominantly matches the platform's US cultural context. Currency is consistently USD with decimal cents across the sample [DATASET-D1, D2, D3, D28, D29], including American-specific features like sales tax [DATASET-D3] and brand-name US candy [DATASET-D1]. US customary units (feet, pounds, square feet) appear regularly [DATASET-D5, D6, D7, D12]. American cultural settings dominate — football, malls, farmer's markets [DATASET-D4, D28]. [WEB-5] confirms 'the standard version is US-centric, using names, currencies, and scenarios typical of Western grade-school curricula.' However, three concerns persist: (a) the paper does not document contractor nationality [Q103, Q104] and the YAML notes a 'transferred from different cultural context' metadata flag (GAP-4); (b) at least one observed problem uses kilograms exclusively [DATASET-D8], conflicting with the platform's US-customary-primary requirement; (c) occasional non-US names and settings appear (DATASET-D14 'castle provisions'; DATASET-D27 British names; DATASET-D15 Scandinavian names). Given MODERATE priority and that the dominant evidence supports US alignment with only minor inconsistencies, score 4 is appropriate.

ID	Evidence
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
D1	Her parents bought a unicorn piñata for \$13 and filled it with all of her favorite treats. They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. — US dollar pricing, brand-name American candy — strongly US cultural context
D2	Linda bought two coloring books at \$4 each, 4 packs of peanuts at \$1.50 each pack, and one stuffed animal. She gave the cashier \$25 and got no change. — USD transactions with decimal cents; cashier context
D3	Austin bought his seven friends each a robot. Each robot costs \$8.75. He was charged \$7.22 total for tax. He left with \$11.53 in change. — USD with tax — American retail context including sales tax
D4	James joins a football team and becomes the star. He scores 4 touchdowns per game and each touchdown is worth 6 points. There are 15 games in the season. He also manages to score 2 point conversions 6 times during the season. — American football scoring system — distinctly US cultural setting
D5	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — US customary units (feet), volume calculation — matches deployment's unit system
D6	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9... the dog reached its final adult weight by adding another 30 pounds — US customary weight (pounds) throughout
D7	he poured into the box enough jelly beans to bring the weight to 2 pounds... he added enough brownies to cause the weight to triple... he added another 2 pounds of jelly beans — US customary weight (pounds) used consistently
D8	Four people lost a total of 103 kilograms of weight. The first person lost 27 kilograms. — Metric weight (kilograms) — not US customary
D12	The price of a home is \$98 per square foot (sq ft). The house is 2,400 sq ft and the barn out back is 1,000 sq ft. How much is this property? — Area (sq ft) applied to pricing — geometry-adjacent; uses sq ft (US customary)
D14	There are enough provisions in a castle to feed 300 people for 90 days. After 30 days, 100 people leave the castle. How many more days are left until all the food runs out? — Multi-step proportional reasoning (rates/ratios) — castle setting, not US-specific

D15	The ratio of coins that Elsa has to that which Amalie has is 10:45. If the total number of coins they have is 440, and Amalie spends 3/4 of what she has on toys — Ratio problem with non-US names; no currency denomination specified
D27	Ian won \$100 in the lottery. He decided to use the money to pay off debts. He paid \$20 to Colin. — USD lottery/debt context — British names (Colin, Helen, Benedict) suggest possible non-US authorship
D28	She went to the mall and spent \$250. Then, she went to the movies and watched 3 movies back to back that each cost \$24. Then she stopped by the farmer's market on her way home and got 20 bags of beans at \$1.25/bag. — USD across multiple retail settings; farmer's market — culturally American context
D29	He sold his \$20,000 car for 80% of its value and then was able to haggle to buy a \$30,000 sticker price car for 90% of its value. — Large-dollar USD transactions, car-buying in American context
WEB-5	Recent cultural adaptation work (Tomar et al. 2025) created non-US variants, confirming original is implicitly US/Western-anchored (https://www.emergentmind.com/topics/gsm8k-dataset-95a37205-4213-4591-900f-09ce82d0a123)

Strengths

- USD currency with decimal cents consistent throughout the sample [DATASET-D1, D2, D3, D28]
- American retail and cultural settings dominate (mall, farmer's market, US football, brand-name candy) [DATASET-D1, D4, D28]
- US customary units (feet, pounds, square feet) used in measurement problems [DATASET-D5, D6, D7, D12]
- [WEB-5] independently confirms US-centric framing as the standard convention

ID	Evidence
D1	Her parents bought a unicorn piñata for \$13 and filled it with all of her favorite treats. They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. — US dollar pricing, brand-name American candy — strongly US cultural context
D2	Linda bought two coloring books at \$4 each, 4 packs of peanuts at \$1.50 each pack, and one stuffed animal. She gave the cashier \$25 and got no change. — USD transactions with decimal cents; cashier context
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D4	James joins a football team and becomes the star. He scores 4 touchdowns per game and each touchdown is worth 6 points. There are 15 games in the season. He also manages to score 2 point conversions 6 times during the season. — American football scoring system — distinctly US cultural setting
D5	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — US customary units (feet), volume calculation — matches deployment's unit system
D6	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9... the dog reached its final adult weight by adding another 30 pounds — US customary weight (pounds) throughout
D7	he poured into the box enough jelly beans to bring the weight to 2 pounds... he added enough brownies to cause the weight to triple... he added another 2 pounds of jelly beans — US customary weight (pounds) used consistently
D12	The price of a home is \$98 per square foot (sq ft). The house is 2,400 sq ft and the barn out back is 1,000 sq ft. How much is this property? — Area (sq ft) applied to pricing — geometry-adjacent; uses sq ft (US customary)
D28	She went to the mall and spent \$250. Then, she went to the movies and watched 3 movies back to back that each cost \$24. Then she stopped by the farmer's market on her way home and got 20 bags of beans at \$1.25/bag. — USD across multiple retail settings; farmer's market — culturally American context
WEB-5	Recent cultural adaptation work (Tomar et al. 2025) created non-US variants, confirming original is implicitly US/Western-anchored (https://www.emergentmind.com/topics/gsm8k-dataset-95a37205-4213-4591-900f-09ce82d0a123)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge

IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — problems contain American cultural references (US football scoring, brand-name candy, mall/farmer's market settings) that benefit from US student familiarity [DATASET-D1, D4, D28]. These are deployment-aligned rather than misaligned.

IC-2: Yes, predominantly. [WEB-5] confirms US-centric framing; empirical sample shows USD throughout (DATASET-D1-D3, D28-D29) and US customary units (DATASET-D5-D7, D12). Minor inconsistencies: one kilogram problem [DATASET-D8] and occasional non-US names [DATASET-D27, D15].

IC-3: No problematic Western-specific knowledge for a US deployment — Western framing IS the target context. Minor non-US-flavored items (castle setting DATASET-D14, British names DATASET-D27) may slightly reduce cultural familiarity but do not impede arithmetic skill assessment.

IC-4: INSUFFICIENT DOCUMENTATION — contractor demographics not disclosed in the paper [Q103, Q104]; no published audit of US-conventions enforcement across the full 8,500 problems. Would require systematic regional annotator review.

IC-5: Minor content issues: ~5-10% of sampled problems show non-US-flavored framing (kilograms, non-US names, non-American settings). The cumulative impact on construct validity is low given that arithmetic skill is unit-agnostic, but reduces 'culturally familiar American settings' fidelity claim modestly.

ID	Evidence
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
D1	Her parents bought a unicorn piñata for \$13 and filled it with all of her favorite treats. They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. — US dollar pricing, brand-name American candy — strongly US cultural context
D2	Linda bought two coloring books at \$4 each, 4 packs of peanuts at \$1.50 each pack, and one stuffed animal. She gave the cashier \$25 and got no change. — USD transactions with decimal cents; cashier context
D3	Austin bought his seven friends each a robot. Each robot costs \$8.75. He was charged \$7.22 total for tax. He left with \$11.53 in change. — USD with tax — American retail context including sales tax
D4	James joins a football team and becomes the star. He scores 4 touchdowns per game and each touchdown is worth 6 points. There are 15 games in the season. He also manages to score 2 point conversions 6 times during the season. — American football scoring system — distinctly US cultural setting
D5	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — US customary units (feet), volume calculation — matches deployment's unit system
D6	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9... the dog reached its final adult weight by adding another 30 pounds — US customary weight (pounds) throughout
D7	he poured into the box enough jelly beans to bring the weight to 2 pounds... he added enough brownies to cause the weight to triple... he added another 2 pounds of jelly beans — US customary weight (pounds) used consistently
D8	Four people lost a total of 103 kilograms of weight. The first person lost 27 kilograms. — Metric weight (kilograms) — not US customary
D12	The price of a home is \$98 per square foot (sq ft). The house is 2,400 sq ft and the barn out back is 1,000 sq ft. How much is this property? — Area (sq ft) applied to pricing — geometry-adjacent; uses sq ft (US customary)
D14	There are enough provisions in a castle to feed 300 people for 90 days. After 30 days, 100 people leave the castle. How many more days are left until all the food runs out? — Multi-step proportional reasoning (rates/ratios) — castle setting, not US-specific
D15	The ratio of coins that Elsa has to that which Amalie has is 10:45. If the total number of coins they have is 440, and Amalie spends 3/4 of what she has on toys — Ratio problem with non-US names; no currency denomination specified
D27	Ian won \$100 in the lottery. He decided to use the money to pay off debts. He paid \$20 to Colin. — USD lottery/debt context — British names (Colin, Helen, Benedict) suggest possible non-US authorship

D28	She went to the mall and spent \$250. Then, she went to the movies and watched 3 movies back to back that each cost \$24. Then she stopped by the farmer's market on her way home and got 20 bags of beans at \$1.25/bag. — USD across multiple retail settings; farmer's market — culturally American context
D29	He sold his \$20,000 car for 80% of its value and then was able to haggle to buy a \$30,000 sticker price car for 90% of its value. — Large-dollar USD transactions, car-buying in American context
WEB-5	Recent cultural adaptation work (Tomar et al. 2025) created non-US variants, confirming original is implicitly US/Western-anchored (https://www.emergentmind.com/topics/gsm8k-dataset-95a37205-4213-4591-900f-09ce82d0a123)

Information Gaps

- Full-dataset frequency of non-US-flavored problems (sample suggests ~5-10% but n=80 too small for precision)
- Whether kilogram or other metric-only problems appear in problem types not sampled
- Contractor nationality and Common Core familiarity

Requires Expert Verification

- Systematic audit by US elementary curriculum specialist of cultural setting and unit-system consistency across the full 8,500-problem corpus
- Confirmation that no benchmark problems use non-USD currency

Remediation

Gap 1: Occasional non-US-flavored problems (kilograms only, non-US names, non-American settings) in roughly 5-10% of sample [DATASET-D8, D14, D15, D27]; contractor nationality undocumented

Recommendation: Filter or flag benchmark items using metric-only units or non-US settings if strict US-context fidelity is required for reporting. Consider using one of the published cultural adaptations or running an automated keyword/unit filter.

ID	Evidence
D8	Four people lost a total of 103 kilograms of weight. The first person lost 27 kilograms. — Metric weight (kilograms) — not US customary
D14	There are enough provisions in a castle to feed 300 people for 90 days. After 30 days, 100 people leave the castle. How many more days are left until all the food runs out? — Multi-step proportional reasoning (rates/ratios) — castle setting, not US-specific
D15	The ratio of coins that Elsa has to that which Amalie has is 10:45. If the total number of coins they have is 440, and Amalie spends 3/4 of what she has on toys — Ratio problem with non-US names; no currency denomination specified
D27	Ian won \$100 in the lottery. He decided to use the money to pay off debts. He paid \$20 to Colin. — USD lottery/debt context — British names (Colin, Helen, Benedict) suggest possible non-US authorship

Input Form (IF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: GSM8K | Context: US K-8 Math Tutoring Platform — Homework Checker Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Full alignment on input form. Both benchmark and deployment are text-only English prose word problems [Q9, Q32], using the Latin alphabet, with no audio, image, or structured-data modality. The empirical sample confirms every problem is plain prose in American English. The platform's input channel (text string submitted via API) matches the benchmark format exactly. No infrastructure constraints, signal distribution differences, or script mismatches exist. This dimension was correctly marked LOWER priority in elicitation.

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)

Strengths

- Text-only English prose matches deployment input channel exactly (Q9, Q32)
- Latin alphabet, US English register match platform requirements
- No modality conversion or preprocessing needed
- Calculator annotations are training-only and transparent to evaluation [Q117, Q119]

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)
Q117	"The calculator annotations were not provided by human contractors: they were generated by a combination of hard-coded logic and a finetuned language model." (p.17)
Q119	"During training, there is no special distinction between the annotated tokens and the rest of the solution: they are all just tokens." (p.17)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both source and target are plain English text strings [Q9, Q32]. No signal-distribution differences.

IF-2: Yes — text-based web/mobile API channel requires no specialized infrastructure beyond standard internet access, which the deployment assumes.

IF-3: No domain-specific form differences. The platform requires text-in/binary-out; the benchmark provides text-in with a clearly delimited final numerical answer.

IF-4: No form mismatches identified.

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)

Output Ontology (OO)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: GSM8K | Context: US K-8 Math Tutoring Platform — Homework Checker Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K's output ontology — binary correct/incorrect judgment determined solely by whether the final numerical answer matches ground truth **[Q31, Q60]** — maps directly onto the deployment's stated requirement of 'binary correct/incorrect judgment on submitted final numerical answer only.' The decision rule is objective and culturally invariant: numerical equality. No categorical labels with cultural value-dependence are involved **[Q60]**. The elicitation explicitly confirms 'an unconventional solution method that yields the correct number is scored as correct,' which matches the benchmark's final-answer-only scoring **[Q60, Q87]**. No taxonomy redesign needed.

ID	Evidence
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
Q87	"This voting process considers only the final answer reached by the individual solutions: the final answer selected is the one with the most votes." (p.11)

Strengths

- Binary correct/incorrect decision rule is structurally identical to deployment requirement **[Q31, Q60]**
- Decision boundary is objective numerical equality — culturally invariant
- Sample confirms clean '##### N' final-answer delimiter format **[DATASET-D13, D21, D22]**
- No partial-credit or step-scoring assumptions that would need to be stripped out

ID	Evidence
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Multi-step four-operations problem across sequential days
D21	She gives away 4/5 of the cupcakes to her classmates. Of the remaining 1/5 of cupcakes, she eats 3 cupcakes. — Fraction operations on a whole — grades 4-5 fraction content
D22	Ten friends decide to get an end-of-year gift for their teacher... four of the group drop out. The remaining friends split the cost equally... each share is now \$8 more — Algebraic reasoning with cost-splitting

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance

OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Single binary label (correct/incorrect on final numerical answer) directly matches platform requirement [Q31, Q60].

OO-2: No missing regional categories — the output space is a number, not a label set requiring regional taxonomy.

OO-3: No culturally encoded categories — numerical equality is the sole decision rule [Q60].

OO-4: No taxonomy redesign needed; benchmark structure already matches deployment.

OO-5: No taxonomy issues identified.

ID	Evidence
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)

Output Content (OC)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: GSM8K | Context: US K-8 Math Tutoring Platform — Homework Checker Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth labels are numerical answers verified by contractor re-solving with a documented 1.7% disagreement rate [Q107, Q108], which the authors acknowledge may understate the true subtle-error rate [Q109]. Because correctness is arithmetically verifiable, regional stakeholder disagreement is structurally minimal — a correct sum is correct regardless of cultural perspective. The dataset sample confirms label cleanliness in most cases [DATASET-D13, D21, D22], but DATASET-D23 illustrates the documented error pattern: 'every sixth customer' in the problem text, while the solution divides by 5 to reach label=10, an ambiguity that could cause a correctly-reasoning model to be marked wrong. This is consistent with the paper's documented 1.7% rate. The calculator annotation imperfection [Q118, Q123] is acknowledged but affects training/evaluation infrastructure rather than ground-truth labels. Contractor nationality remains undocumented [Q105], but for a numerical-equality decision rule, this matters less than for subjective tasks. Score 4 reflects strong label quality with documented minor error rate and no cultural-disagreement risk.

ID	Evidence
Q105	"After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q108	"We estimate this to be the fraction of problems that contain breaking errors or ambiguities." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)
Q118	"The logic for auto-generating calculator annotations is imperfect. It is highly unlikely to generate any incorrect annotations, but it is not uncommon for it to ignore some lines that could be annotated." (p.17)
Q123	"We note that the original version of our calculator, used for all results in this paper, had some minor implementation bugs." (p.17)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Multi-step four-operations problem across sequential days
D21	She gives away 4/5 of the cupcakes to her classmates. Of the remaining 1/5 of cupcakes, she eats 3 cupcakes. — Fraction operations on a whole — grades 4-5 fraction content
D22	Ten friends decide to get an end-of-year gift for their teacher... four of the group drop out. The remaining friends split the cost equally... each share is now \$8 more — Algebraic reasoning with cost-splitting
D23	Dan owns an ice cream shop and every sixth customer gets a free ice cream cone. Cones cost \$2 each. If he sold \$100 worth of cones, how many free ones did he give away? — Division with a logical twist — note: answer logic has arithmetic error in solution (50/5=10, but every 6th, not 5th)

Strengths

- Labels are arithmetically verifiable; correctness is objective [Q60]
- Two-pass contractor re-solving validation with documented agreement rate [Q105, Q106, Q107]
- Sample confirms clean numerical labels in most examples [DATASET-D13, D21, D22]
- 1.7% disagreement rate is low enough for a screening-tier benchmark [Q107]

ID	Evidence
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Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
Q105	"After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote." (p.15)
Q106	"We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Multi-step four-operations problem across sequential days
D21	She gives away 4/5 of the cupcakes to her classmates. Of the remaining 1/5 of cupcakes, she eats 3 cupcakes. — Fraction operations on a whole — grades 4-5 fraction content
D22	Ten friends decide to get an end-of-year gift for their teacher... four of the group drop out. The remaining friends split the cost equally... each share is now \$8 more — Algebraic reasoning with cost-splitting

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels are numerical and arithmetically verifiable, so regional stakeholder agreement is essentially structural rather than perspectival.

OC-2: Minimal potential disagreement — US Common Core teachers would converge on the same numerical answers as the original annotators for well-formed problems. Disagreement risk is concentrated in the ~1.7% of problems with breaking errors or ambiguities [Q107, Q108].

OC-3: INSUFFICIENT DOCUMENTATION — contractor demographics (nationality, native language, CCSS-M familiarity) not disclosed [Q103, Q104, Q105]. For a numerical-equality task this is lower-stakes than for subjective labeling.

OC-4: Re-annotation by US K-8 teachers would likely affect <2% of items; not necessary for the deployment but useful for high-stakes use.

OC-5: No aggregation method that could erase minority perspectives — labels are single numerical values, not aggregated judgments.

OC-6: Documented 1.7% disagreement rate [Q107] and acknowledged 'larger percentage may contain subtle errors' [Q109]; one such ambiguity observed empirically [DATASET-D23]. Acceptable for screening-tier evaluation; users should be aware that 1-2% of failures may reflect label issues rather than model errors.

ID	Evidence
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
Q105	"After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote." (p.15)

Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q108	"We estimate this to be the fraction of problems that contain breaking errors or ambiguities." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)
D23	Dan owns an ice cream shop and every sixth customer gets a free ice cream cone. Cones cost \$2 each. If he sold \$100 worth of cones, how many free ones did he give away? — Division with a logical twist — note: answer logic has arithmetic error in solution (50/5=10, but every 6th, not 5th)

Information Gaps

- True subtle-error rate beyond the documented 1.7% breaking-error rate
- Contractor demographics and CCSS-M familiarity

Requires Expert Verification

- Targeted re-annotation of a sample by US elementary teachers to estimate CCSS-M-aligned label disagreement rate

Remediation

Gap 1: Documented 1.7% label-disagreement rate [Q107] and acknowledged subtle-error possibility [Q109]; one ambiguity observed empirically [DATASET-D23]

Recommendation: Have US K-8 math teachers re-validate a sample of GSM8K items used in user-facing accuracy reporting, or restrict reported metrics to a curated subset where label quality has been re-confirmed.

ID	Evidence
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)
D23	Dan owns an ice cream shop and every sixth customer gets a free ice cream cone. Cones cost \$2 each. If he sold \$100 worth of cones, how many free ones did he give away? — Division with a logical twist — note: answer logic has arithmetic error in solution (50/5=10, but every 6th, not 5th)

Output Form (OF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: GSM8K | Context: US K-8 Math Tutoring Platform — Homework Checker Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Full alignment. GSM8K’s output form is a final numerical answer string evaluated via exact-match correctness [Q29, Q46], which maps directly onto the deployment’s stated output_channel ('binary correct/incorrect label returned to platform UI') and answer_format ('single numerical value — integer, decimal, or fraction'). The dataset sample confirms clean '#### N' final-answer delimiters across all examined examples [DATASET-D13, D21, D22]. No text-to-speech, multi-modal output, or accessibility transformations are needed. Literacy is presumed in the target K-8 student population reading word problems in English. No format mismatch exists.

ID	Evidence
Q29	“At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct.” (p.5)
Q46	“Test performance is determined by a single low temperature ($T = 0$) sample for each test problem.” (p.6)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Multi-step four-operations problem across sequential days
D21	She gives away $\frac{4}{5}$ of the cupcakes to her classmates. Of the remaining $\frac{1}{5}$ of cupcakes, she eats 3 cupcakes. — Fraction operations on a whole — grades 4-5 fraction content
D22	Ten friends decide to get an end-of-year gift for their teacher... four of the group drop out. The remaining friends split the cost equally... each share is now \$8 more — Algebraic reasoning with cost-splitting

Strengths

- Final numerical answer with exact-match scoring matches deployment output schema exactly [Q29, Q46]
- Clean '#### N' delimiter format across the sample [DATASET-D13, D21, D22]
- No partial-credit or step-scoring that would need to be discarded
- Text output channel matches platform UI requirement

ID	Evidence
Q29	“At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct.” (p.5)
Q46	“Test performance is determined by a single low temperature ($T = 0$) sample for each test problem.” (p.6)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Multi-step four-operations problem across sequential days
D21	She gives away $\frac{4}{5}$ of the cupcakes to her classmates. Of the remaining $\frac{1}{5}$ of cupcakes, she eats 3 cupcakes. — Fraction operations on a whole — grades 4-5 fraction content
D22	Ten friends decide to get an end-of-year gift for their teacher... four of the group drop out. The remaining friends split the cost equally... each share is now \$8 more — Algebraic reasoning with cost-splitting

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Text-based numerical output matches platform UI requirement (binary label returned via API).

OF-2: Not applicable — deployment is text-based, not speech-based.

OF-3: Target population (K-8 students using English-language platform) is assumed literate in US English; no accessibility transformation needed for benchmark evaluation purposes.

OF-4: No form mismatches identified.

ID	Evidence
Q29	"At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct." (p.5)
Q46	"Test performance is determined by a single low temperature ($T = 0$) sample for each test problem." (p.6)